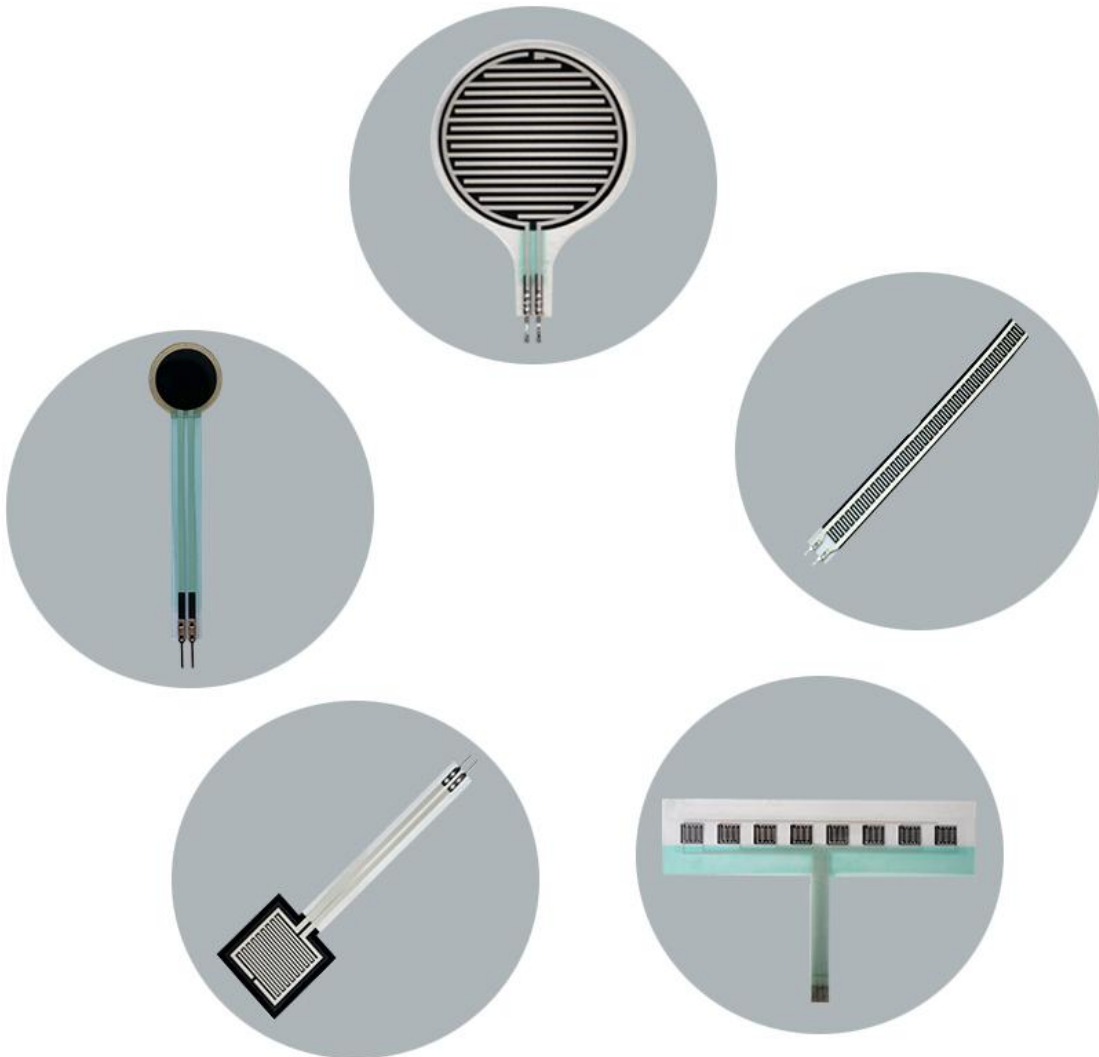


Membrane Force Sensor Catalogue



Description :

Flexible pressure sensor (also called FSR) is an ultra-thin (thickness usually around 0.3mm), high-sensitivity resistive pressure sensor. When pressure is applied to the sensing area, the resistance of the sensor changes. The greater the pressure, the lower the resistance. At the same time it will cause the output voltage to change, the greater the pressure, the greater the output voltage. This kind of sensor is mainly used to measure the pressure change trend in the area, or use multiple sensors to place different locations to measure the pressure distribution.

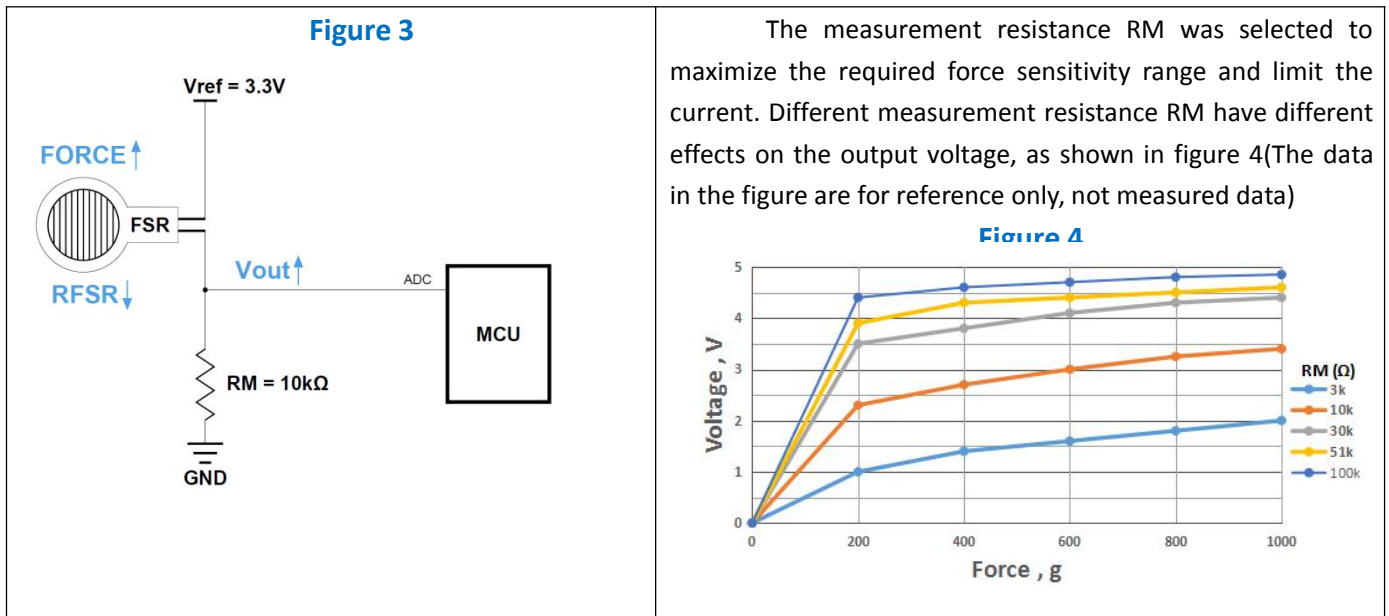
There are many applications for this pressure sensor, such as robotic grip sensing, human and animal gait measurement, wheelchair sitting measurement, electronic musical instruments, smart boxing gloves, pressure measurement insoles and more. However, because the pressure detection is not very accurate, it is not recommended to use the situation where the accurate pressure detection is required.

Integration and usage methods:

According to the characteristic that FSR resistance value is inversely proportional to the pressure applied on its surface, a basic circuit connection diagram is provided in figure 3 (the selection of Vref and RM values in the figure is a reference value, which can be adjusted by the user according to the actual situation). We put FSR and shunt resistor RM in series in the circuit, apply Vref reference voltage at both ends of the circuit, connect the ADC pin of MCU between FSR and RM, and use the ADC function of MCU to detect the Vout voltage value.

The calculation formula of the output voltage is as follows:

$$R_{FSR} = \frac{RM}{V_{out}} (V_{ref} - V_{out})$$





Model	Image	Sensing area	Force range
FA101		$\Phi 3.8\text{mm}$	100g ~ 10KG
FA201		$\Phi 9.5\text{mm}$	100g ~ 10KG
FA301		$\Phi 10\text{mm}$	100g ~ 10KG

FA400		Φ 15.7mm	100g ~ 10KG
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Model	Image	Sensing area	Force range
FA401-A		Φ 26mm	100g ~ 10KG
FA401-B		Φ 26mm	100g ~ 15KG

FA402		Φ 26mm	100g ~ 10KG
FA408		160*12mm	200g ~ 20KG

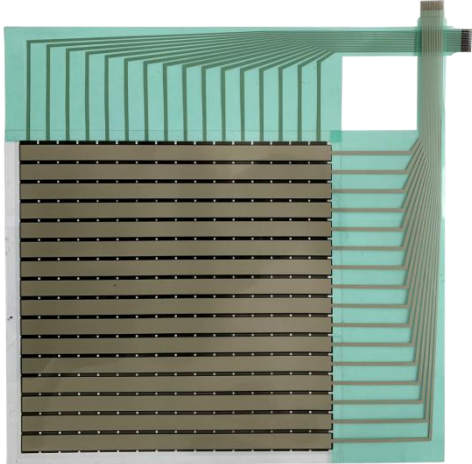


Model	Image	Sensing area	Force range
FA502-A		51.40*50.8mm	100g ~ 10KG

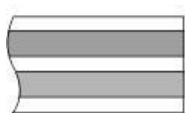
FA502-B		51.40*51.40mm	100g ~ 10KG
FSR F01		Φ 38.5mm	100g ~ 10KG
FSR F02		19.05*19.68mm	100g ~ 15KG



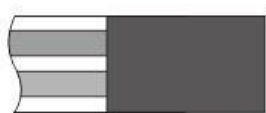
Model	Image	Sensing area	Force range
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Matrix FSR 16*16		229.78*229.78mm	200G ~ 50KG
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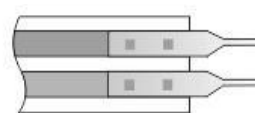
Connector Information



Bare Tail



Female connector



Male connector

Customization

We accept orders for customized force sensor. If you have such need, please contact with our sales team. If you have any technical questions about force sensing resistor, please feel free to email our engineer: thomson@flexkys.com.